

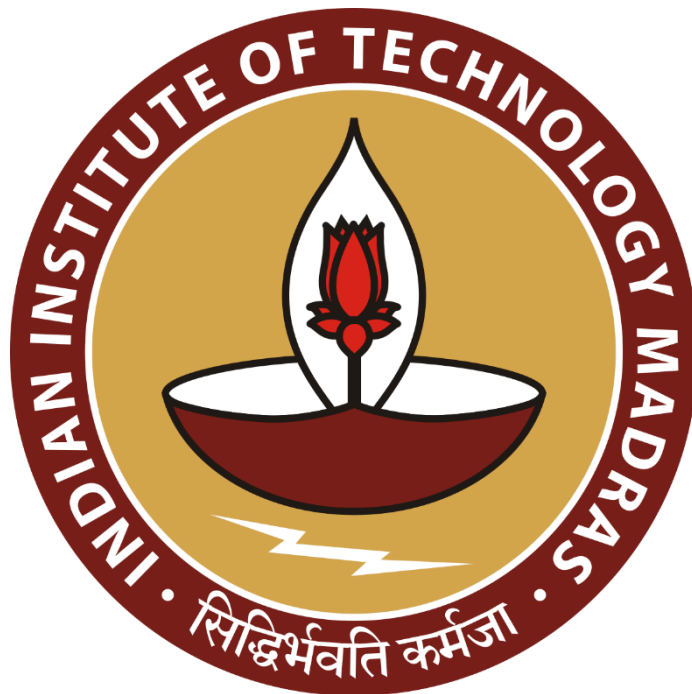
The Seasonal Sales and Demand Trends In Sahitya Sadan Bookstore

A Final Term report for the BDM capstone Project

Submitted by

Name : Aman Kumar Mahali

Roll number : 23f2000459



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1.	Executive Summary.....	2
2.	Detailed Explanation of Analysis Process.....	3
3.	Results and Findings.....	8
4.	Interpretation of Results and Recommendation.....	16

1. Executive Summary

The Sahitya Sadan is a small bookstore established by Mr. Durga Shankar's father back in 1972, the bookstore is located in the Musabani market, East Singhbhum district, Jharkhand. The bookstore works on a B2C (Business-to-Consumer) model and serves all the academic / non academic essentials like academic books, Notebooks, Stationeries, Office essentials, art materials etc. and is operated by Mr. Jagmohan Patar for 7 years now. Due to the diverse range of products it becomes difficult to track the availability of all the products, and this leads to the inefficiencies such as overstocking for low demand items and stockouts for some items whose demand is currently high in the market, on a particular period of time, the business is also facing an issue of customers loyalty, because some customers gave orders for some items in bulk, but they did not come back to purchase the same due to some reasons.

To address the issues (inventory management and customer loyalty), 3-months of sales data (1st April, 2024 to 30th June, 2024) was collected from the bookstore. Since, the bookstore uses the application named "My BillBook", in which all the transactions are being recorded in a digital format, the raw data was obtained from the business owner and then it was cleaned first for the analysis, and to address the challenges the bookstore is facing. The data-driven analysis was done in the Google Sheets, which includes the following : Descriptive Analysis, ABC Analysis, Monthly transaction Analysis, Monthly Sales Analysis, Analysis of the revenue generated in different days of a week, Analysis of revenue generated from different Books, finding the Top 5-Fast/Slow selling items, Economic Order Quantity Analysis, Safety Stocks Analysis, Reorder Point Analysis and at last Recency Frequency Monetary Analysis.

From the analysis of the sales data, it was found that the Books category contributes 62.8% of the total revenue generated in 3-months. The monthly transaction patterns clearly highlighted that, the sales of the Books are seasonal as it dips after the academic season gets over. The monthly sales trend highlighted the month April, as it generated the highest revenue of all 3-months. The revenue generated in different days of a week highlighted the influence of the market days. The revenue generated from different books highlighted the dominance of the JCERT (Jharkhand Board) books. The top 5 fast selling/slow selling items proves that these items are low cost consumables and also confirms the brand's dominance. The inventory analysis highlighted the volatile demand and bulk purchases of some items. The highest vs lowest profit analysis proves that the exam guides generate the maximum profits. At last, the RFM analysis clearly identified 3 loyal customers so far.

Based on the quantitative analysis of the data and a qualitative analysis through discussions with the owner, a detailed set of recommendations has been provided which includes a strong communication with the supplier for faster delivery, use of EOQ, Safety Stocks and Reorder Point data to better manage the inventory and use of RFM score to identify the loyal customers. This helped the bookstore to reduce the chances of both stock-outs and the over stocking of items as well as reduced the customers loyalty issue. The above measures also helped the Sahitya Sadan Bookstore to improve the profitability and enhance the customers relationships, which ultimately helped for the growth of the business.

2. Detailed Explanation of Analysis Process

2.1 Data Cleaning and Preprocessing :

The main problem which Sahitya Sadan bookstore is facing, is the inventory management issues (stockouts for the high-demand items and the overstocking of the low-demand items), hence from 1st April, 2024 to 30th June, 2024 (3 months) of the sales data has been collected from the bookstore. The Sahitya Sadan bookstore uses the “My BillBook” application, in which each and every sales transaction has been recorded in a raw format, then the raw data was extracted from the application and transferred into a Google Sheets for the cleaning and the analysis of the same.

The [Raw Dataset](#), which was extracted from the “My BillBook” application contains the followings 10 columns : **Invoice No, Invoice Date, Contact Name, Amount, Remaining Amount, Invoice Status, Due Date, Invoice Link, Payment Type, Created by**, etc. and total 3107 rows. Since, the name of actual items, quantity of items, MRP of the items was missing from the raw data and this information was present in the invoice’s link. Hence, all the invoice’s links generated during the 3 months were opened, and the data (Items, Quantity, MRP, Total) present inside it was extracted into another new Google Sheets. During the extraction of the data from the invoice’s links, the challenges like : automatic addition of extra columns, data redundancy, etc. arises, which was carefully managed during the data entry time.

For the better analysis of the data, the un-necessary columns which was present in the raw data like : Invoice No, Contact Name, Amount, Remaining Amount, Invoice Status, Due Date, Invoice Link, Payment Type, Created by, etc was not written in the new Google Sheets, and instead of that new columns like : **Serial No., Items, Quantity, MRP, Total, Category, Month, Day**, etc. was written along with the **Invoice Date** in the new Google Sheets, which can be accessed by this link : [Cleaned Dataset](#). All these steps played a critical role in the data analysis process and ensured the data quality as well as the accurate analysis of the data.

The Sahitya Sadan bookstore sells a lot of items, hence it was necessary to categorize all the items based on the following : Stationaries (like pen, pencil, fevicol, files, cello tapes, etc.), Books (like NCERT, JCERT, guide books, G.K., dictionary, etc.), Paper Goods (like Notebooks, registers, forms, diary, maps, etc.) and Art materials (like colors, drawing copy, brush, coloring books, etc.). And to categorize the same, the category column was created using the function : `=IF(OR(ISNUMBER(SEARCH("Item_name", Column number))), "Category"`

2.2 Comprehensive Explanation :

Once the data was ready, the analysis of the same was started in the Google sheets. The first step taken was to create a pivot table in the Google Sheets, because that will help to calculate, summarize and analyze the patterns and trends in the 3-months of sales data. The below is the same, which was created using the sales data.

Month	Art materials	Books	Paper Goods	Stationary	Monthly Total
April	₹3,375.00	₹2,05,880.00	₹91,001.10	₹7,499.00	₹3,07,755.10
May	₹895.00	₹75,944.00	₹39,466.00	₹6,861.00	₹1,23,166.00
June	₹1,017.00	₹42,296.00	₹31,490.00	₹10,474.00	₹85,277.00
Grand Total	₹5,287.00	₹3,24,120.00	₹1,61,957.10	₹24,834.00	₹5,16,198.10

Image_1 : Pivot Table

From the pivot table it was clearly seen that among the 3-months, the highest revenue was generated in the month of April (**₹3,07,755**), followed by May (**₹1,23,166**) and then June (**₹85,277**). To further break down the information gained from the pivot table, the descriptive analysis of the data was done.

2.2.1 Descriptive Analysis :

The descriptive analysis was the second step taken towards the data analysis process because through this analysis we can understand the fundamental characteristics of the sales data and it also summarizes the total 4,983 individual transactions into a clear and interpretable story. The calculations of the key statistical measures of central tendency (Mean, Median, Mode) and the measure of variability (Standard deviation, Range) was done using their respective formulas.

2.2.2 ABC Analysis :

The ABC analysis was the third step taken towards the data analysis process, since the Sahitya Sadan bookstore is facing an inventory management issues, it was important to classify the items into 3 categories :

- **Category A** includes the high-demand products, or items which are less in number but make up almost 80% of total revenue.
- **Category B** includes the mid-range products, or items which are more in number than Category A and make up almost 15% of the total revenue.
- **Category C** includes low-demand products, or items which are highest in quantity but low in value and make up almost 5% of the total revenue.

For this analysis, the Items and the Total column was cleaned first, just to ensure that the Items column had consistent names (no extra spaces, duplicates due to spelling) and the Total column contained only numeric values. Then to aggregate the sales data by the unique Item name, a pivot table was created in the new Google Sheets, then the items were sorted in descending order of the revenue. After this, the cumulative percentage was calculated for the total revenue. Based on the cumulative percentage, items contributing to the first 80% of total revenue were classified as Category A, the next 15% (up to 95%) as Category B, and the remaining 5% as Category C. The ABC analysis revealed that the top 76 items accounted for approximately 80% of the revenue, the next 99 items contributed around 15%, and the remaining 172 items contributed only 5%, demonstrating the Pareto principle in sales distribution.

Hence, the ABC analysis was done through the steps mentioned above, which helped the bookstore to carefully manage the inventory, categorize and focus on the items which drives the majority of the revenue and will ultimately lead to increase in the profits of the business.

2.2.3 Monthly transaction Analysis :

The Monthly transaction analysis was the fourth step taken towards the data analysis process, because it'll help to identify the transaction patterns seen during the 3 months. The dates were first normalized and converted into a "Month" attribute (April, May, June), since the raw sales data contained individual transaction dates (e.g., "01/04/2024"), this allowed the aggregation of daily transaction noise into clear, comparable monthly periods, enabling the identification of broader seasonal patterns. Then a pivot table was created using the column month, category and the corresponding transactions count. Then to visualize the monthly transaction pattern, a line chart (Chart_2 : Monthly transactions) was plotted which gave a clear and interpretable pattern from the data. The pattern seen during the 3-months will help the Sahitya Sadan bookstore to be prepared for these 3 upcoming months (April to June), and carefully manage the inventory.

2.2.4 Monthly Sales Analysis :

Similar to the Monthly transaction analysis, the Monthly sales analysis was the fifth step taken towards the data analysis process and to conduct the same, the pivot table was constructed first using the Month, Category and the corresponding Total column, after that to visualize the Monthly sales trend, a Column chart (Chart_3 : Monthly sales trend by category) was plotted using the pivot table. This analysis tracked how the monetary contribution of each product category evolved from the peak season (April) to the off-season (June), and also tracked how customer demand and sales shifts across different product categories, this will help the Sahitya Sadan Bookstore to be prepared on those 3-months and carefully manage all the inventory as per the demand of the market, and reduce the chances of Stock-outs of high demand items and Over-stocking of low demand items.

After this analysis, the Top-5 highest vs. Lowest profitable products were also plotted in Chart-9.

2.2.5 Revenue in different days of a week :

The analysis of the revenue generated in the different days of a week was the sixth step taken towards the data analysis process, for this the corresponding days are written first, then a pivot table was created using the Days and the Total column, and to visualize the same a Column chart (Chart_4 : Total revenue Vs day of the week) was plotted. This analysis helped the Bookstore to be prepared and meet the demands of the market on the days in which the sales were maximum (market days).

2.2.6 Revenue generated from Books :

The analysis of the revenue generated from different books was the seventh step taken towards the data analysis process, since it was clear that the Books category drives the majority of the revenue. First the dataset was filtered to exclude all non-book categories, this created a clean subset of data containing only Book transactions, then new attribute was created for the Book category like : JCERT (Jharkhand Board), SM Series (Private Publisher), NCERT (Central Board) and Others. This step transformed thousands of unique book titles into four clear and strategic groups. Then a pivot table was created using Book category and Total column to aggregate the financial performance of these newly defined groups. To visualize the same, a Pie chart (Chart_5 : Pie chart showing revenue generated w.r.t. Book type) was plotted.

2.2.7 Top 5-Fast/Slow Selling items :

The analysis of the Top 5-Fast/Slow selling items was the eighth step taken towards the data analysis process, for this the Quantity column was cleaned first, as it contained text formats, then for the aggregation of the data a pivot table was constructed using Item and Quantity column, this aggregated data was sorted in descending order based on the Quantity, this gave the top 5-Fast selling items. For the top 5-slow selling item, the aggregated data was sorted in the ascending order. A Column chart (Chart_6 : Top 5-Fast selling items and Chart_7 : Top 5-Slow selling items) was plotted to visualize the same.

2.2.8 EOQ (Economic Order Quantity) Analysis :

Finding the EOQ of the Top 5-Fast selling items was the ninth step taken towards the data analysis process. First, the Items name, Total Quantity Sold, Projected Demand (D), Unit Cost, Holding Cost per unit (H), Ordering Cost (S), etc. was written on the Google Sheets for the top 5 Fast selling items, and then the EOQ was calculated using the below formula :

$$EOQ = \sqrt{\frac{2DS}{H}}$$

For calculations, Ordering cost is assumed to be ₹100 per order (transport, phone calls, labor) and holding cost per unit per year is assumed to be 20% of the item's cost per year (storage space, damage risk).

2.2.9 Safety Stocks Analysis :

Finding the Safety Stocks was the tenth step taken towards the data analysis process, first a pivot table was created using the Invoice Date and Sum of Quantity, this generated a chronological list. Then the maximum Daily Sales and the Average Daily Sales was calculated, followed by the Lead time (the time between placing an order and receiving it). For calculations, the Max. Lead time and Avg. Lead Time is 10 days, 7 days respectively. After calculating all the values, the Safety Stocks of Top 5-Fast selling items was calculated using the below formula :

$$\text{SS} = (\text{Max. Daily Sales} \times \text{Max. Lead Time}) - (\text{Avg. Daily Sales} \times \text{Avg. Lead Time})$$

2.2.10 ROP (Reorder Point) Analysis :

Finding the ROP was the eleventh step taken towards the data analysis process, since the Average Daily Sales, Average Lead Time (for calculations, it is taken to be 7 days) and the Safety Stock was already calculated in the previous step (Safety Stock), then all the values was used to calculate the Reorder Point using the below formula :

$$\text{ROP} = (\text{Average daily Sales} \times \text{Avg. Lead time}) + \text{Safety Stock}$$

2.2.11 RFM (Recency, Frequency, Monetary) Analysis :

To deal with the problem (Lack of Customer Loyalty), Mr. Jagmohan was requested to maintain a Logbook in which the Month name along with the Date (For Recency), Customer ID (For unique identification), Customer Name (For reference), Bill Amount (For Monetary), Visit count (For Frequency), etc. was maintained and then all the data was converted into a digital format (Google Sheets) for the analysis. A pivot table was created to group the data by Customer ID, then Recency (Day since last visit), Frequency (Total visits) and the Monetary (Total Spent) was calculated and based on it, the following Score was assigned as : 1 (For Risk), 2 (For Average) and 3 (For Best). This analysis helped the business to identify their loyal customers.

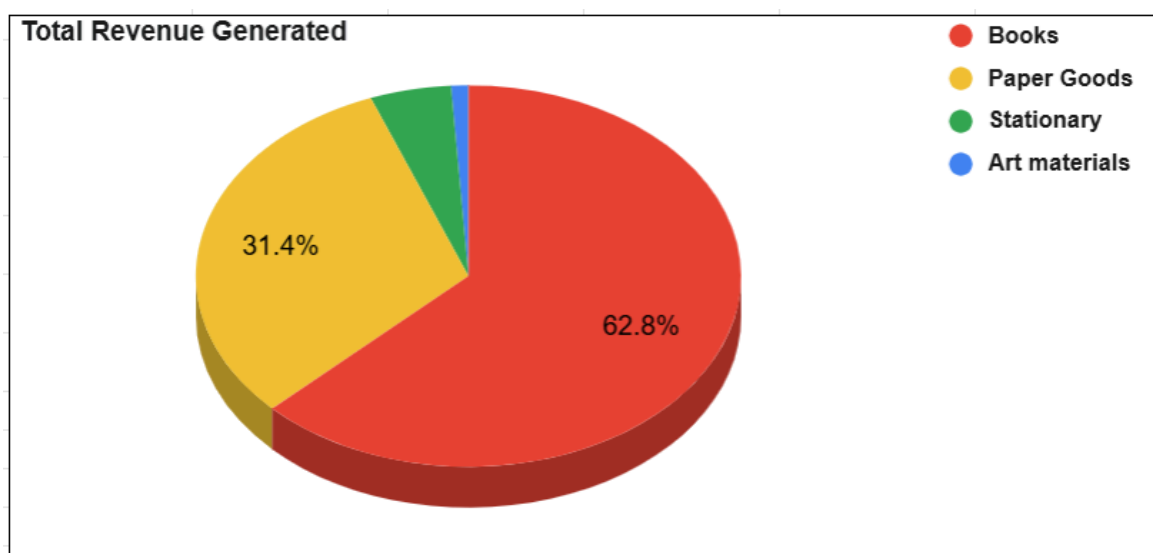
2.2.12 Method to reduce the Wastage :

As mentioned in the report that some items get wasted due to pests, hence some of the methods like : no inventory should be kept directly on the floor, improvement of storage arrangement by using closed containers for loose items, sealed wall gaps, use of repellents, etc. have been implemented to deal with this problem and reduce the wastage of the inventory.

3. Results and Findings

After the detailed analysis of the 3-months sales data, the results and findings are mentioned below :

3.1 ABC Analysis :



Chart_1 : Pie chart showing Category wise Total Revenue

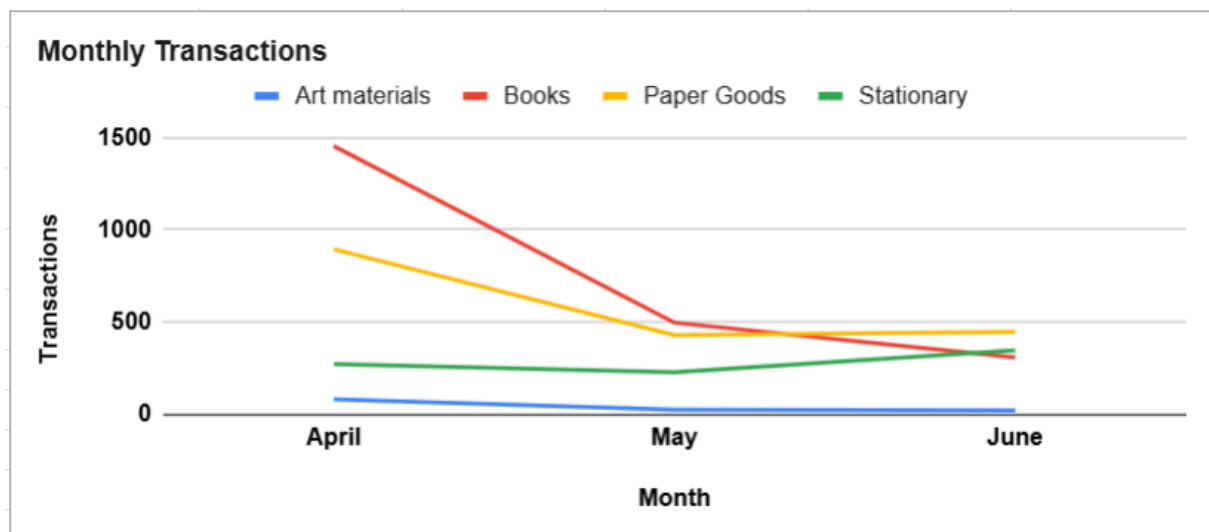
The ABC analysis of the data can be clearly seen through the pie chart above, the “Books” category contributes Rs. 3,24,120 (62.8%) followed by the “Paper Goods” category with Rs. 1,61,957 (31.4%), then “Stationary” category with Rs. 24,834 (4.8%) and at last “Art Materials” category with Rs. 5,287 (1%) of the total revenue during the 3-months.

The **Category A** items are less in number but still they drive the majority of the revenue. Some of the products which came under this category are the following : JCERT(Jharkhand Council of Educational Research and Training) Books from primary to Secondary classes, Nalanda Long Exercise Books, Saathi Rule 164 Pages A4 Size Copy, Amar prime 120 Pgs rule notebook, 176 pgs Amar prime rule notebook, etc.

The **Category B** items are mid-range products, or items which are more in number than Category A and generate less revenue per item than Category A. Some of the products which came under this category are the following : Amar Drawing Books, Doms Instruments Box, Saathi A4size Notebook, Amar bound register no 16, ALKA Passport books, etc.

The **Category C** items include low-demand products, or items which are highest in quantity but low in value, and they are necessary for footfall but don’t drive the major revenue. Some of the products which came under this category are the following : Apsara pencil, pens, Extra long Eraser(Rubber), Colors, chart papers, sheets, etc.

3.2 Monthly Transaction Patterns :



Chart_2 : Monthly transactions

The above graph clearly shows the monthly transactions pattern seen during the 3-months (April to June).

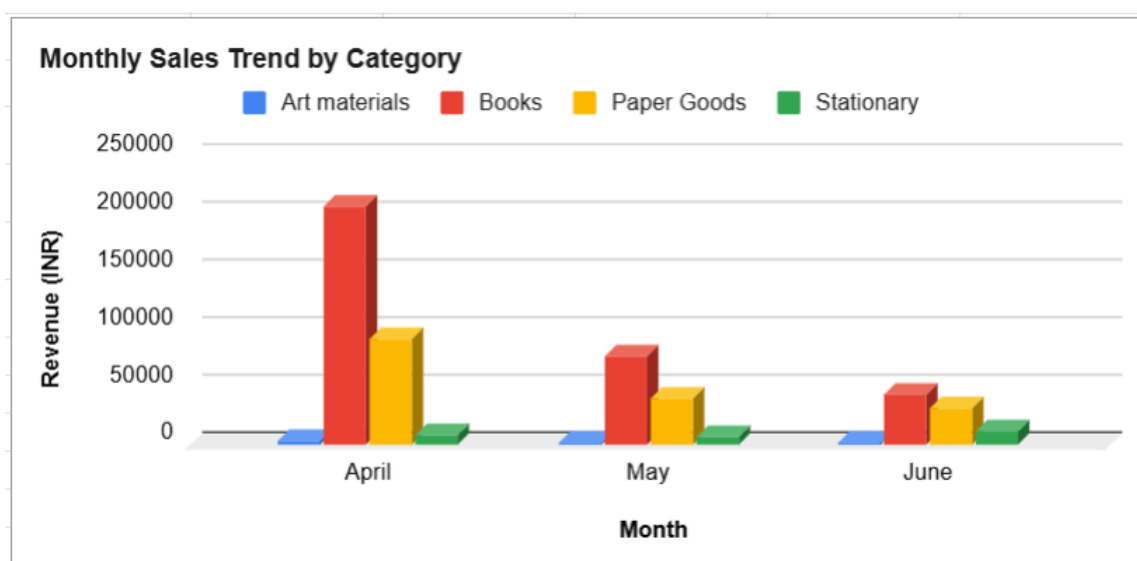
The **Books** category is purely driven by the academic calendar, as it showed a massive spike in April (1,455 transactions) followed by a sharp crash in May (495 transactions) and June (306 transactions). All these transaction patterns seen in this month, is due to the “back-to-school” rush because in JAC (Jharkhand Academic Council) board, the academic session starts from the month of April and once a customer buys their set, they do not return for this category which confirms that the sales of the book are one-time event per year.

The **Paper Goods** were highest sold in April (894 transactions), and then dipped in May (428 transactions), but slightly recovered in June (445 transactions). The Paper Goods have a recurring demand, as students fill up notebooks and need more. The slight rise in June suggests that the first wave of replenishment is already beginning as schools likely fully resumed or homework loads increased.

The **Stationary** sales were almost stable from April (270 transactions) to May (225 transactions) and then rose slightly in June (345 transactions). While customers stopped buying expensive books, they've started visiting frequently for small, low-value items like pens and replacements, the footfall is now driven by Stationary, not Books. This also confirms that Stationary items are not seasonal, they are consistent.

The **Art Materials** are often part of the initial school list (buying a color box in April), hence the sales were highest in April (79 transactions) and then dipped in May (22 transactions) and again dipped in June (19 transactions), because unlike pens or notebooks, they last a long time, hence the flatline in sales after the initial rush.

3.3 The Monthly Sales Trend :



Chart_3 : Monthly sales trend by category

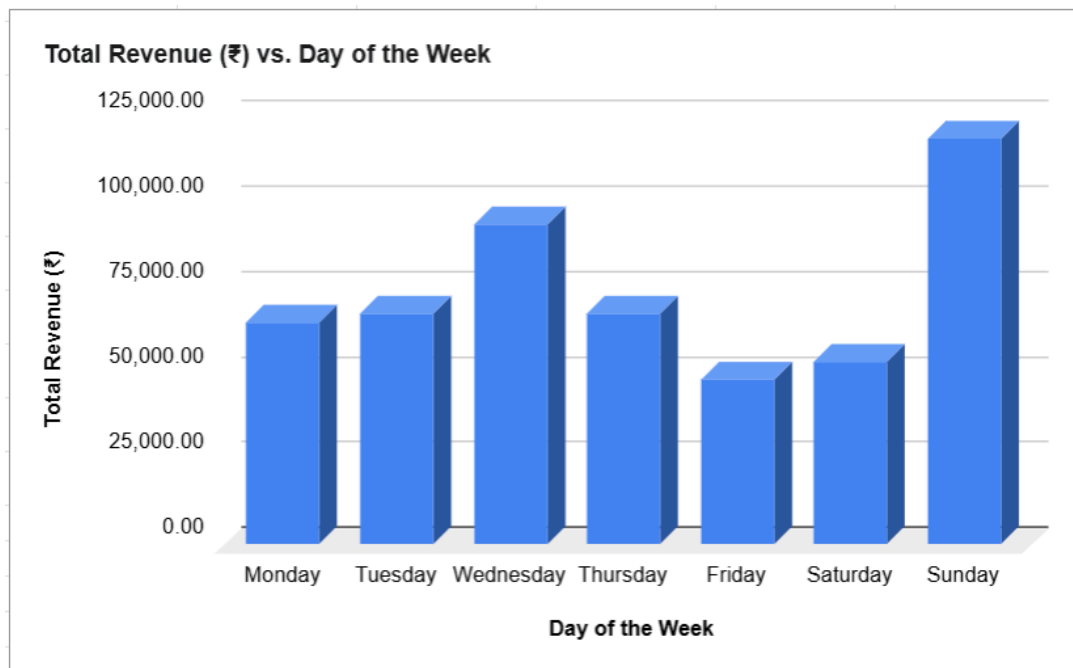
The above graph provides a clear visualization of the revenue trends seen during these 3-months across all four categories.

In **April**, the total revenue generated was ₹3,07,755 (the highest of all three months), this is due to the “back-to-school” rush (as the academic session starts from April in JAC board), and customers made their first academic purchase. The Books are the primary revenue driver because, total ₹2,05,880 revenue was generated from this category only, followed by ₹91,001 revenue from Paper Goods category and ₹7,499 revenue from Stationary category and at last ₹3,375 was generated from Art Materials category.

In **May**, the total revenue generated was ₹1,23,166, it significantly dropped (from ₹3,07,755 to ₹123,166), the Books category still remained the primary revenue driver as ₹75,944 revenue was generated from this category, followed by ₹39,466 revenue was generated from Paper Goods and ₹6,861 revenue was generated from Stationary category and at last ₹895 was generated from Art Materials category. The revenue drop seen in this month is because the customers had already bought their supplies in April (as the academic session starts from April in JAC board), while the amount of money dropped, the behaviour stays somewhat similar to April (buying sets), just at a much lower volume.

In **June**, the total revenue generated was ₹85,277 (the lowest of all three months), because the academic season was over and the daily retail season began. ₹42,296 revenue was generated from the Books category, followed by ₹31,490 from Paper Goods and ₹10,474 from Stationary and at last ₹1,017 from Art Materials category. Stationary is the only category that performed better in June than in May or April, maybe because the customers might have lost their first pen or finished their first pencil, or need a geometry box for a new chapter. The sales of Books dropped to their lowest point, and the sales are likely limited to a specific reference books, guides, or lost textbook replacements.

3.4 Revenue generated in different days of a week :



Chart_4 : Total revenue Vs day of the week

The above column chart clearly shows the “twin-peak” revenue trend with respect to the different days of the week.

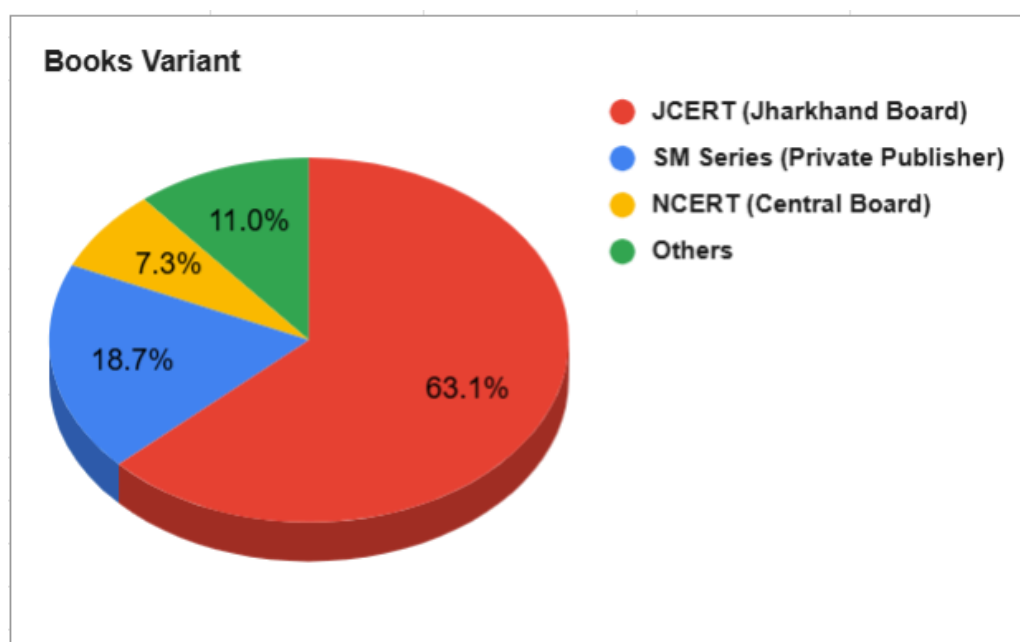
Sunday is the primary peak, as it generated ₹1,18,967 (the highest of the week) which is nearly double the revenue of typical weekdays like Monday or Friday. This is because buying a full set of academic books/items requires time and money, and working parents are free on Sundays to accompany their children.

On **Monday and Tuesday**, the revenue generated was almost the same ₹65,257 and ₹67,932 respectively, this clearly showed that the footfall reduces after the peak seen on Sunday.

Wednesday is the secondary peak, a total ₹93,817 (the second highest of the week) was generated, this is because Wednesday is the weekly market day and customers might be combining their grocery shopping with buying items from the bookstore.

On **Thursday, Friday and Saturday**, the revenue generated was ₹67,964, ₹48,460 (the lowest of the week) and ₹53,801 respectively. From the column chart it can be clearly seen that the revenue dropped on Thursday and it attained the minimum value on Friday (₹48,460) and again it rose on Saturday (from ₹48,460 to ₹53,801), the pattern seen on these days is maybe because these days are not the market days and many customers preferred to buy items on the market days. Due to these reasons the revenue declined from Thursday and attained the minimum on Friday and it again rose on Saturday.

3.5 Revenue generated from Books :



Chart_5 : Pie chart showing revenue generated w.r.t. Book type

From the Chart_1 (Pie chart which showed the category wise total revenue), it was clear that the books are the primary revenue drivers, hence it was necessary to find out the books which were sold the most during the academic season.

From the above Pie chart (Chart_5), it's clear that the bookstore is fundamentally powered by the "JCERT (Jharkhand Board)" books (63.1%) followed by "SM Series (Private Publisher)" books (18.7%), then "NCERT (Central Board)" books (7.3%), and at last "Others" books (11%).

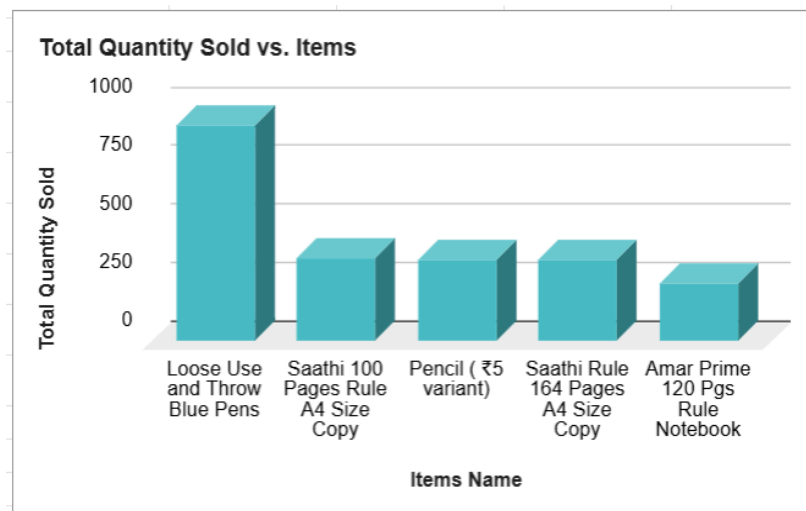
The **JCERT** books contribute ₹2,04,445 (63.1%) of the total revenue generated from the Books category. This clearly shows that JCERT books are the most demanded books during the academic season, and some of them are the following : Devvani Sanskritam 2 JCERT, Abha Prabhat Honey Comb- 7 JCERT, Surbhi Samanya Gyan 7 JCERT, Science and Consciousness 4 JCERT, etc.

The **SM Series (Private Publisher)** books contribute ₹60,465 (18.7%) of the total revenue generated from the Books category, and some of them are the following : SM Kala Pujan 2, SM My Computer 1, SM Creative English Reader 1, SM Prachi Mathematics 4, etc.

The **NCERT (Central Board)** books contribute ₹23,710 (7.3%) of the total revenue generated from the Books category, and some of them are the following : Mathematics Class 7 NCERT, NCERT Physics For 11, Mathematics Class 6 NCERT, etc.

The **Others** books contribute ₹35,500 (11%) of the total revenue generated from the Books category, and some of them are the following : Bhargava's Pocket Dictionary English-Hindi, GO GRAMMAR 7, APNA VYAKARAN, etc.

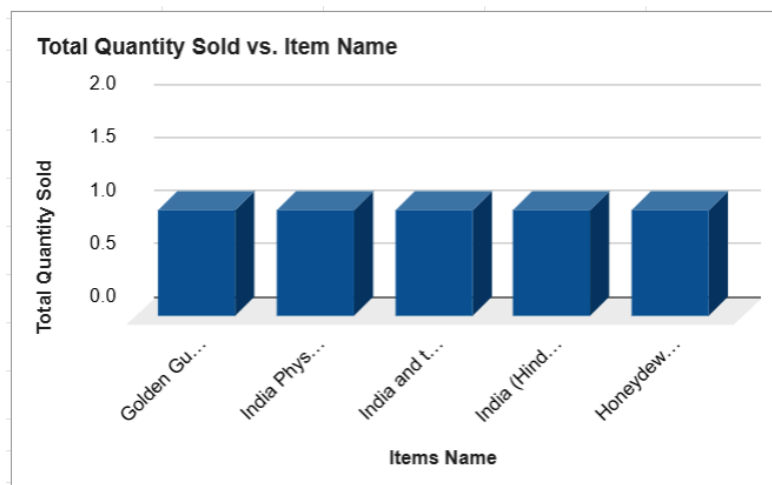
3.6 Top 5-Fast Selling items :



Chart_6 : Top 5-Fast selling items

The “Loose Use and Throw Blue Pens” (924 quantity) are the most sold item during the 3-months, followed by “Saathi 100 Pages Rule A4 Size Copy” (350 quantity), then “Pencils” (348 quantity), then Saathi Rule 164 Pages A4 Size Copy (344 quantity) and at last Amar Prime 120 Pgs Rule Notebook (246 quantity). This proves that Top selling items are low-cost consumables, also Saathi A4 notebooks appeared twice in the list, confirming the brand’s dominance.

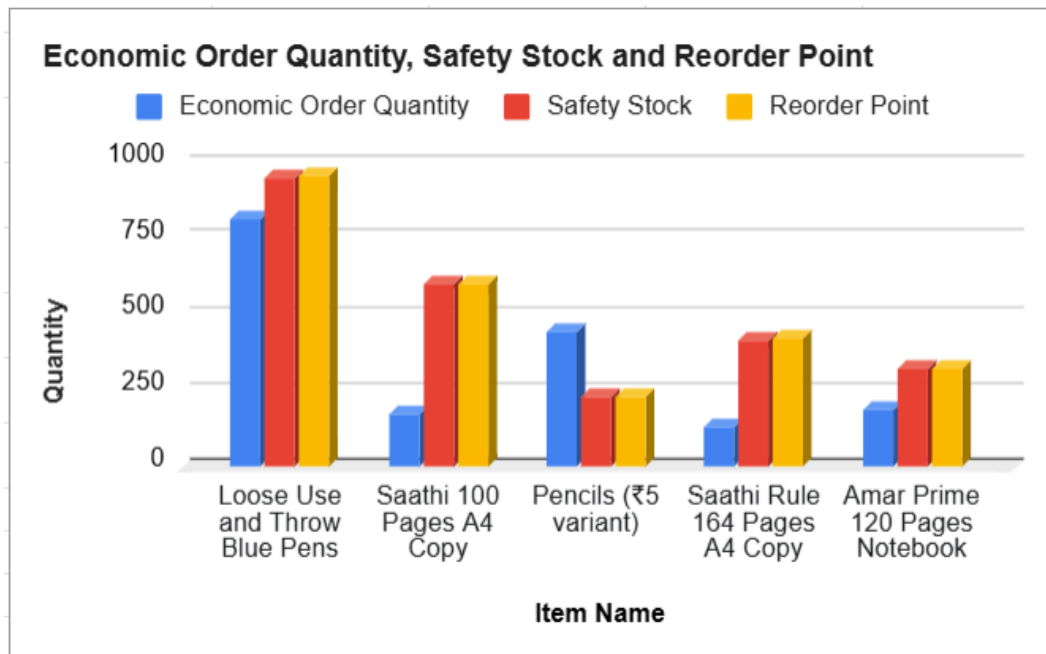
3.7 Top 5-Slow Selling items :



Chart_7 : Top 5-Slow selling items

The “Golden Guide Book Chapterwise Class-10”, “India Physical & Adjacent Countries Map”, “India and the Contemporary World-2 (Class 10)”, “India (Hindi) Map Tinpatti” and “Honeydew Class 8 NCERT”, was sold only 1 quantity in 3-months, making these the top least selling items.

3.8 Inventory Analysis (Economic Order Quantity, Safety Stocks, Reorder Point) :



Chart_8 : Inventory Analysis

For **Loose Use and Throw Blue Pens**, the Economic Order Quantity is 815 units, while the Reorder Point is 960 units and the Safety Stock is 951 units, both are nearly identical. The high Safety Stock indicates that the demand is highly volatile, likely driven by unpredictable bulk purchases.

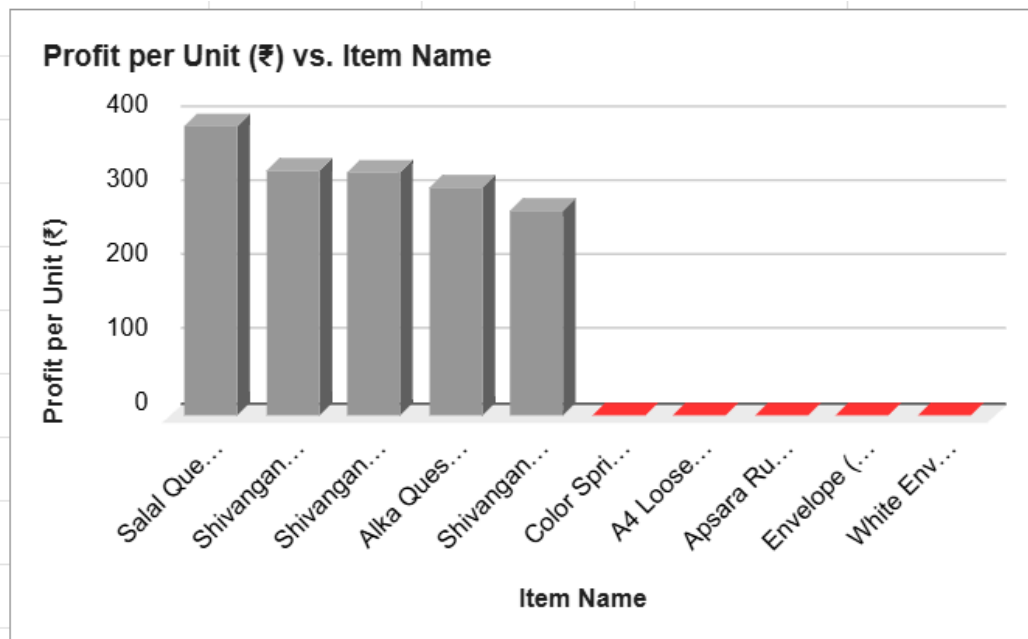
For **Saathi 100 Pages A4 Copy**, the Economic Order Quantity is 172 units, and the Reorder Point is 600 units which is nearly 3.5 times higher than the Economic Order Quantity. The Safety Stock is 597 units. The high Reorder Point indicates that to protect against massive demand surge, the bookstore has to order the items even if the inventory levels are high.

For **Pencils (₹5 variant)**, the Economic Order Quantity is 446 units, which is nearly double the Reorder Point (230 units), and the Safety Stock is 227 units. The overstocking of this item is strategically viable due to its low unit cost and non-perishable nature

For **Saathi Rule 164 Pages A4 Copy**, the Economic Order Quantity is 132 units which is significantly lower than the Reorder Point (420 units). The Safety Stock is 417 units, indicating that the inventory levels are inflated by seasonal bulk outliers.

For **Amar Prime 120 Pages Notebook**, the Economic Order Quantity is 189 units, and the Reorder Point (320 units) is almost entirely driven by the Safety Stock (318 units). The negligible difference between the ROP and Safety Stock implies that while average daily sales may be moderate, the item likely faces supply chain reliability issues (long or uncertain delivery times).

3.9 Highest vs. Lowest Profit Analysis :



Chart_9 : Highest vs. Lowest Profit

The Top-5 high profit items are the followings : Salal Question Bank 2025 Class 12 with a profit of ₹390, Shivangan J.T.E.T Teacher Guide (Primary) with a profit of ₹330, Shivangan J.T.E.T Teacher Guide (Social) with a profit of ₹325, Alka Question Bank 2026 Arts Class 12 with a profit of ₹306 and Shivangan Joint Competition Exam with a profit of ₹275. This clearly proves that the Exam Guides generate the maximum profit (between ₹275 to ₹390), which the owner also highlighted in the [interaction video](#). These 5 items act as the “Financial Anchors” of the store, where selling just a few quantities generates more profit than selling thousands of low profit items.

The Top-5 low profit items are the followings : Color Sprint Copier Paper (Per Sheet), A4 Loose Paper (Per Sheet), Apsara Rub-A-Long Eraser, Envelope (Small), White Envelope (Medium) with a profit of ₹0.20, ₹0.25, ₹0.25, ₹0.40, ₹0.40 respectively. These items just create the ecosystem that brings customers into a store.

3.10 RFM Analysis :

The Recency, Frequency, Monetary Analysis of the month (December) highlighted the total 241 unique customers and only 3 loyal customers (Sita Besra, Ratan Tudu and Sujit Tudu) have been identified so far. Sita visited 3 times in a month and spent ₹857, Ratan and Sujit also visited 3 times and spent ₹1,035, ₹1,555 (Highest spender in the list) respectively. Only these three customers demonstrate repetitive purchase behavior and others are just one time traffic.

4. Interpretation of Results and Recommendation

4.1 Inventory Management Issues :

- **Interpretation :**

From the above section, it's clear that the sales of some items are seasonal, and it highly depends on the academic season. So, over-stocking of the items (like books, Paper Goods) after the academic season does not contribute much in the total revenue, hence it's better to carefully manage the specific category after the academic season.

Also, during the academic season, it's necessary to have a good quantity of academic items (like academic books, paper goods, etc.), since most customers buy items in the beginning of the academic season (April), and stock-outs of any item during this time may lead to the loss in the business.

4.1.1 ABC Analysis :

The ABC analysis clearly distributed the items into 3-category (A, B, and C) based on the quantity sold and the revenue generated from the same. This analysis highlighted that 62.8% of the total revenue was generated from the Books category, also the majority revenue of the Bookstore was driven by the Books category.

4.1.2 Monthly Transaction Patterns :

The Monthly transaction also highlighted that the Books category is purely driven by the academic calendar and showed a massive spike in April.

4.1.3 The Monthly Sales Trend :

The Monthly sales trend highlighted the month April, as ₹3,07,755 (the highest of all three months) revenue was generated in this month.

4.1.4 Revenue generated in different days of a week :

The revenue generated in different days of a week clearly highlighted the influence of market days as the sales were relatively higher on Wednesdays and Sundays.

4.1.5 Revenue generated from Books :

The revenue generated from Books clearly highlighted the dominance of JCERT (Jharkhand Board) books, as ₹2,04,445 (63.1%) of the total revenue was generated from the JCERT books.

4.1.6 Top 5-Slow Selling items :

The Top 5-Slow selling items highlighted that the sales of Maps (India Physical & Adjacent Countries, Map Tinpatti) is very less in the Bookstore.

4.1.7 Top 5-Fast Selling items :

The Top 5-Fast selling items highlighted that the fast selling items are low cost consumables and also confirms the brand's dominance, as Saathi A4 notebooks appeared twice in the list.

4.1.8 EOQ (Top 5 Fast-Selling Items) :

For Loose Use and Throw Blue Pens, the Economic Order Quantity is 815 units, since these pens are cheap and small, they cost almost nothing to store, and these items should be bought in bulk (800-900 pens), this might also help to negotiate a discount which ultimately increase the profit margin. Also, the sales of stationery is not seasonal (customers often buy these items every month).

For Pencils (₹5 variant), the Economic Order Quantity is 446 units, it's almost similar to pens but the volume is slightly lower. This item also came under the Stationary category, and the sales are not seasonal, hence an order of 446 pencils covers roughly 7-8 months.

For Saathi 100 Pages A4 Copy, the Economic Order Quantity is 172 units. Notebooks are heavier and take up shelf space and the holding cost is also higher than pens, hence 172 units is a healthy balance and is enough for roughly 3-months of supply.

For Amar Prime 120 Pages Notebook, the Economic Order Quantity is 189 units. This is relatively cheaper to hold and even though the sales are lower than Saathi A4 copy, the EOQ is higher (189 vs 172).

For Saathi Rule 164 Pages A4 Copy, the Economic Order Quantity is 132 units. This should be ordered more frequently in smaller batches compared to the others and 132 units is enough for roughly 2-months of supply.

4.1.9 Safety Stocks (Top 5 Fast-Selling Items) :

For Loose Use and Throw Blue Pens, the Safety Stock is 951 Units. Hence, it's better to keep roughly 1000 pens as safety stock and will be reserved for the bulk buyers.

For Pencils (₹5 variant), the Safety Stock is 227 units. Hence, it's better to keep roughly 230 pencils as safety stock, to meet the demand anytime.

For Saathi 100 Pages A4 Copy, the Safety Stock is 597 units. Hence, it's better to keep roughly 600 units as safety stock, such that even if the supplier is completely out of stock, this will not affect the bookstore.

For Amar Prime 120 Pages Notebook, the Safety Stock is 318 units. Hence, it's better to keep roughly 320 units, because even if the bookstore runs out of Saathi notebooks, this stock will offer customers a cheaper alternative.

For Saathi Rule 164 Pages A4 Copy, the Safety Stock is 417 units. Hence, it's better to keep roughly 420 units as a safety stock, just like the above Saathi notebook.

4.1.10 ROP (Top 5 Fast-Selling Items) :

For Loose Use and Throw Blue Pens, the Reorder Point is 960 units. Hence, the bookstore should order this item if the stock dips below 1000 units roughly.

For Pencils (₹5 variant), the Reorder Point is 230 units. Hence, the bookstore should order this item if the stock dips below 230 units roughly.

For Saathi 100 Pages A4 Copy, the Reorder Point is 600 units. Hence, the bookstore should order this notebook if the stock dips below 600 units.

For Amar Prime 120 Pages Notebook, the Reorder Point is 320 units. Hence, the bookstore should order this notebook if the stock dips below 320 units. This also acts as a backup brand if the Saathi brand notebooks run out of stocks.

For Saathi Rule 164 Pages A4 Copy, the Reorder Point is 420 units. Hence, the bookstore should order this notebook if the stock dips below 420 units.

- **Recommendations :**

For the **academic season**, the bookstore should procure 100% of the projected demand by March 25th, and maintain a Safety Stock of 10 days peak volume throughout April. The revenue generated in the first week of April should be used to fund the second batch of procurement. By preventing the stockouts of any item in the academic season will directly impact the profits of the business because the sales of some categories of items are highly dependent on the academic season.

On **market days** (like wednesday and sunday), the staff should be ready to handle all the customers because the footfall is maximum on these days, and is the best time to generate more revenue from the business. One important thing to notice before the market days is the availability of the inventory, since stockouts of any item may impact the profits of the business. Hence, by providing the faster service on the market days will help the business to generate more revenue.

For the **top 5 fast-Selling items** (like Loose Use and Throw Blue Pens, Pencils (₹5 variant), Saathi 100 Pages A4 Copy, Amar Prime 120 Pages Notebook and Saathi Rule 164 Pages A4 Copy, etc.) the bookstore should pay more attention on the number of stocks left on the bookstore and once the items quantity dips below the reorder point, they should place an order for the same.

For the **faster delivery** of the goods, the communication should be strong with the suppliers such that the delivery of the goods may not get delayed during the academic/non-academic season. This will ensure that the bookstore is ready to match the expected demand.

For the **non academic season**, the bookstore should focus on the other categories like Paper Goods, Stationary, etc. because after the end of the academic season, the revenue will not be driven by the Books category, and Paper Goods, Stationary may take a lead as customers often buy these categories of items.

4.2 Lack of Customer Loyalty :

- **Interpretation :**

The RFM clearly highlighted the three loyal customers so far, and if the RFM score is low for some customer then they should avoid accepting an order of items if advance payment is not done.

Also, the Lack of Customers Loyalty is an Operational Reliability problem and may arise due to the following reasons :

Stock reliability failure : if the customer is a bulk buyer and the bookstore was unable to fulfill the remaining 50% of the demand due to some reason (stockouts), then it may happen that from next time, the customer moves to another bookstore who offers one-stop-reliability.

Time-to-Value friction : if the customer had given order of some item, but the bookstore take 5-7 days to procure the same, then customers often solves the problem elsewhere like borrow from other, buys a second hand or they may choose to buy the item from another bookstore, because customers always wants the items as soon as possible, etc.

- **Recommendations :**

Enforce a mandatory 50% advance payment for any item not currently in stock, for the customers whose RFM score is not good.

Zero-Deposit Privilege to customers whose RFM score is good and has a proven track record. Waiving the deposit acts as a perk, making them feel trusted and valued, which deepens their commitment to the Bookstore.